Human Anatomy

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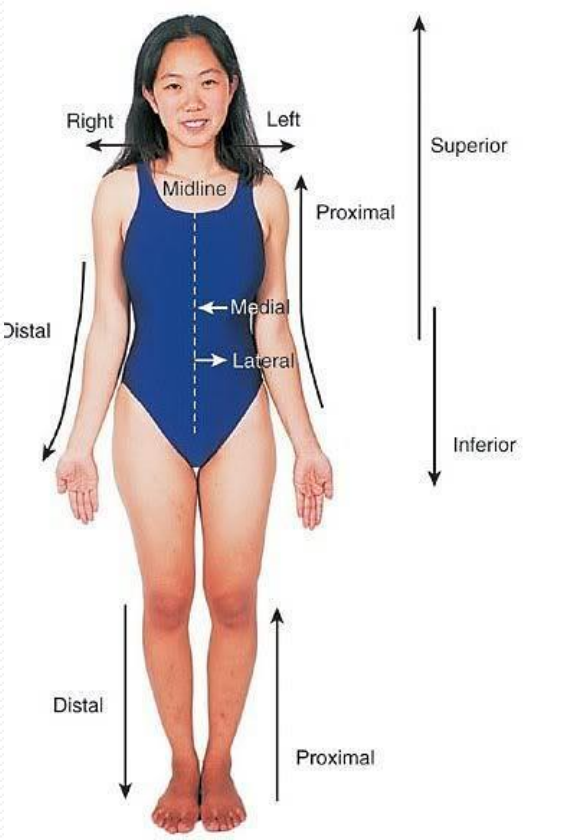
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Part I

Systemic Anatomy

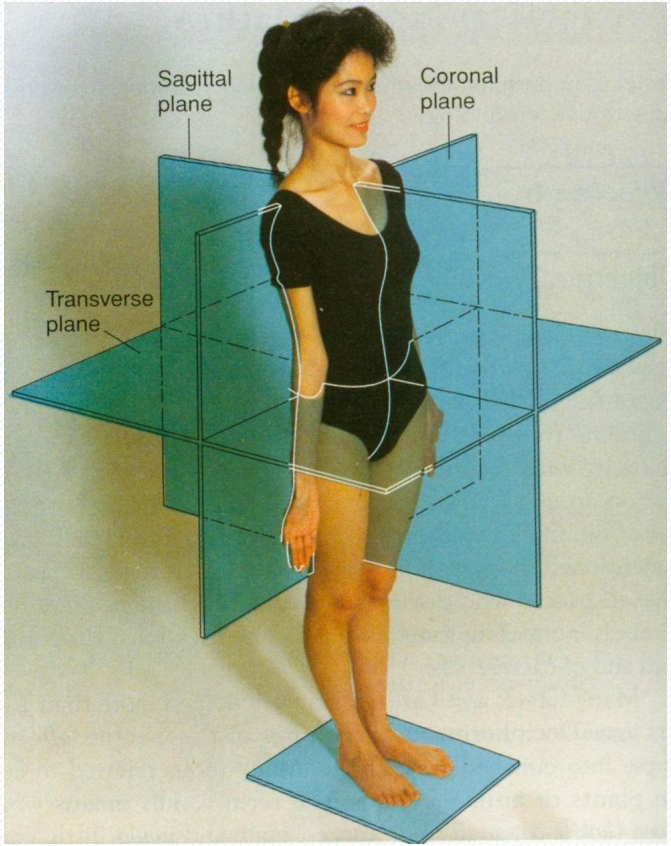
Anatomical position

Human body, in the anatomical sense, is described as standing position with eyes looking forward, upper limbs hanging on each side of the body, palms facing forward, heels getting together and big toes facing forward



Axis and Planes

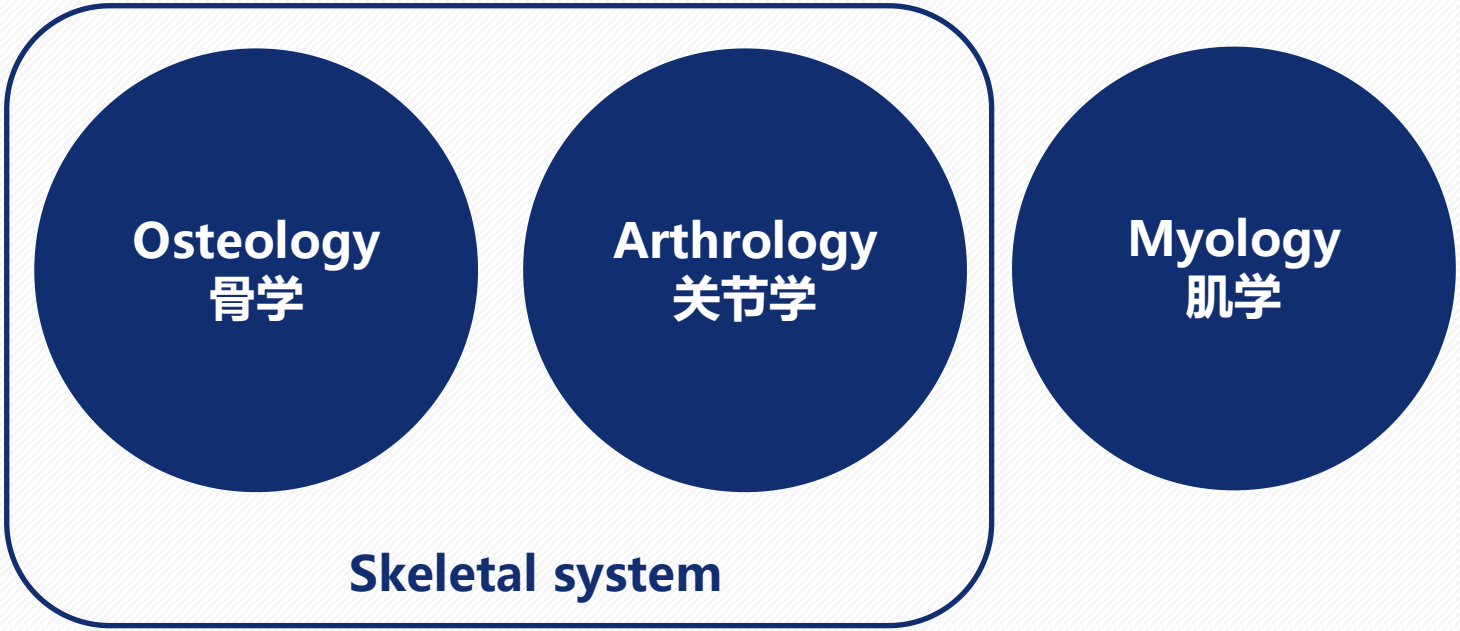
- **Axis**
  - Vertical axis 垂直轴
  - Sagittal axis 矢状轴
  - Coronal axis 冠状轴
- **Plane**
  - Sagittal plane 矢状面
  - Coronal plane 冠状面
  - Horizontal plane 水平面



Systemic Anatomy

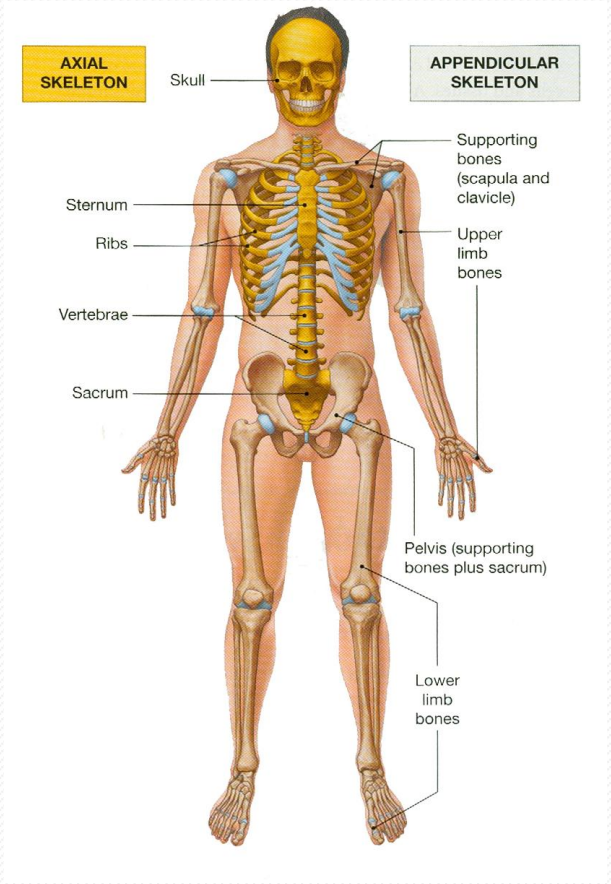
- Locomotor system 运动系统
- Digestive system 消化系统
- Respiratory system 呼吸系统
- Urinary system 泌尿系统
- Reproductive system 生殖系统
- Cardiovascular system 心血管系统
- Lymphatic system 淋巴系统
- Sensory organs 感觉器
- Nervous system 神经系统
- Endocrine system 内分泌系统

Locomotor system



Skeletal system

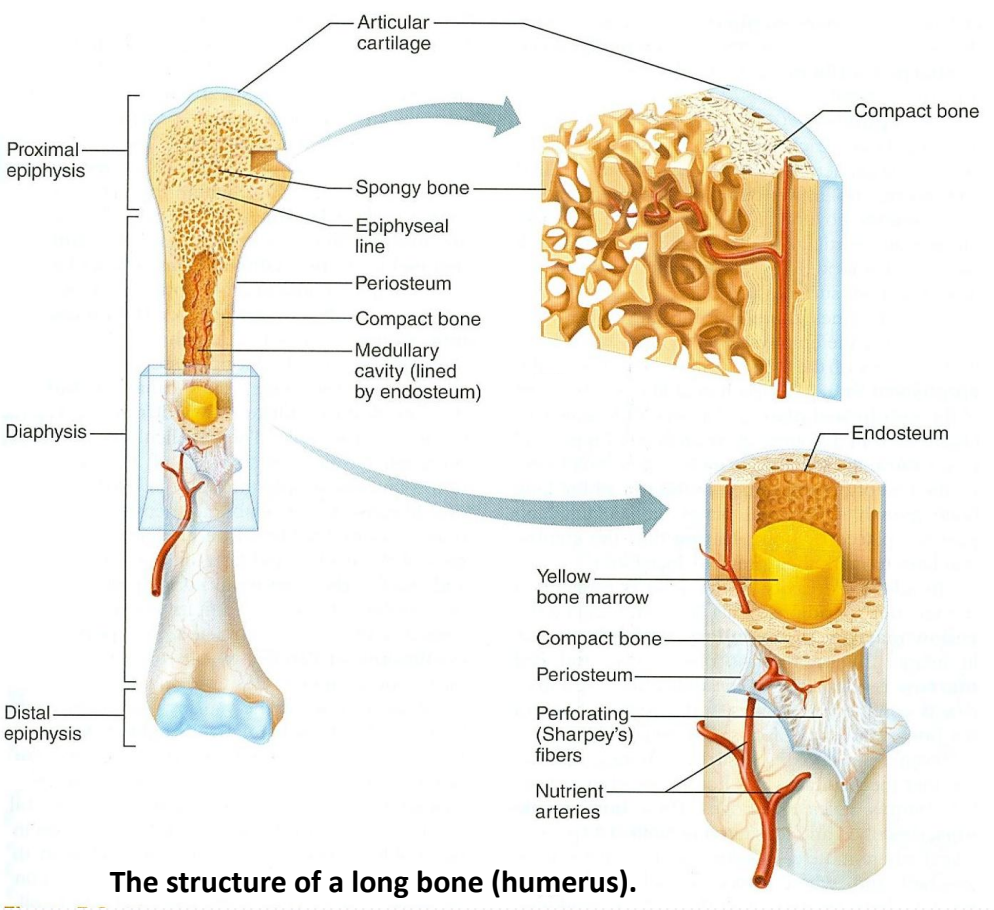
The skeletal system is composed of the **bones, cartilages,** and **joints** that form the internal framework.



Bones

Structure of Bone

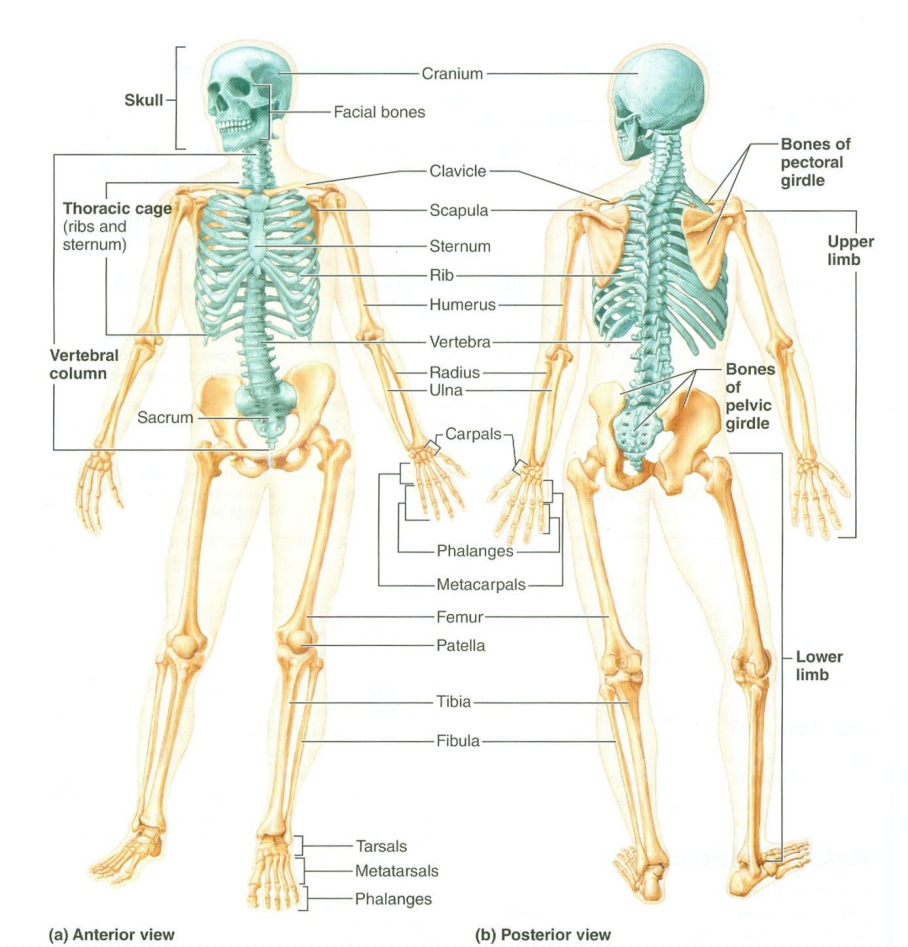
- Compact bone 骨密质
  Spongy bone 骨松质
- Periosteum 骨（外）膜
  Endosteum 骨内膜
- Bone marrow
- Blood vessels and nerves



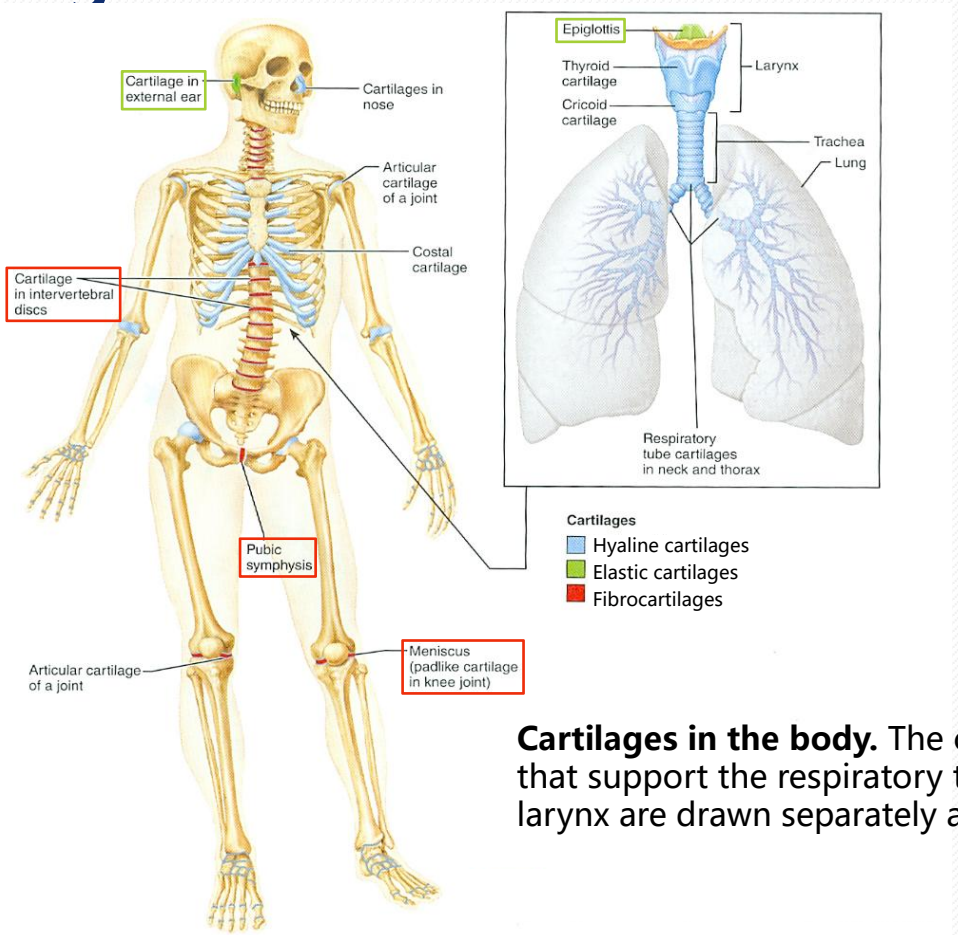
The structure of a long bone (humerus).

Bones

The human skeleton.
The bones of the axial skeleton are colored green. Bones of the appendicular skeleton are gold.



Cartilages

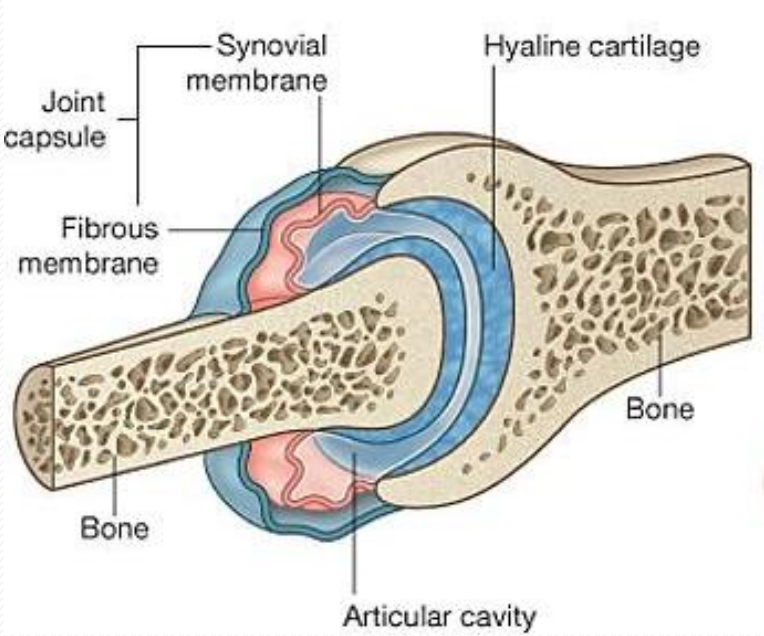


Cartilages in the body. The cartilages that support the respiratory tubes and larynx are drawn separately at right.

Joints

General Structure of synovial joints

- Articular surface
- Articular capsule
- Articular cavity



Major features of a synovial joint.

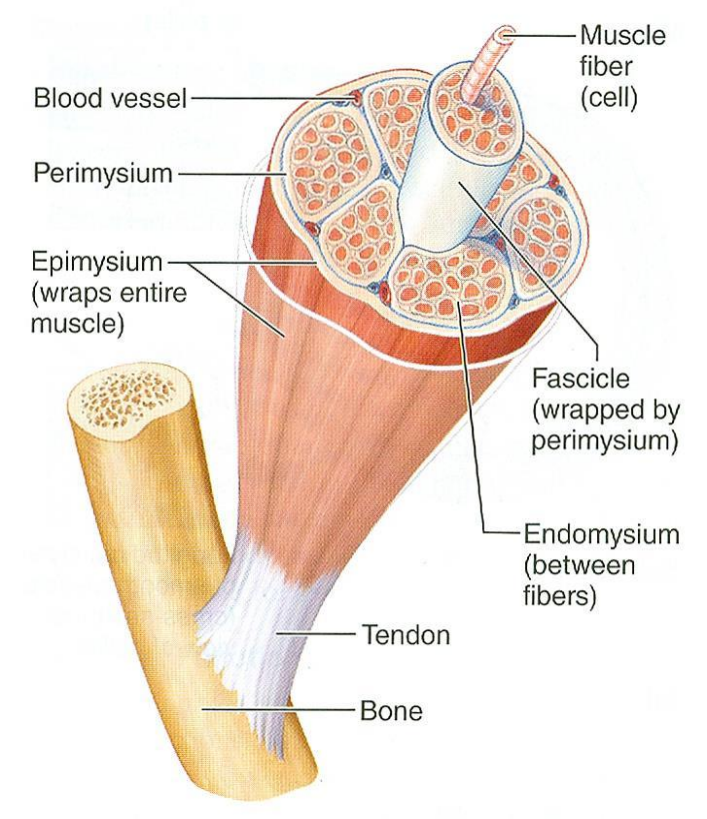
Myology

Skeletal muscle

- Muscle belly
- Tendon

Gross anatomy

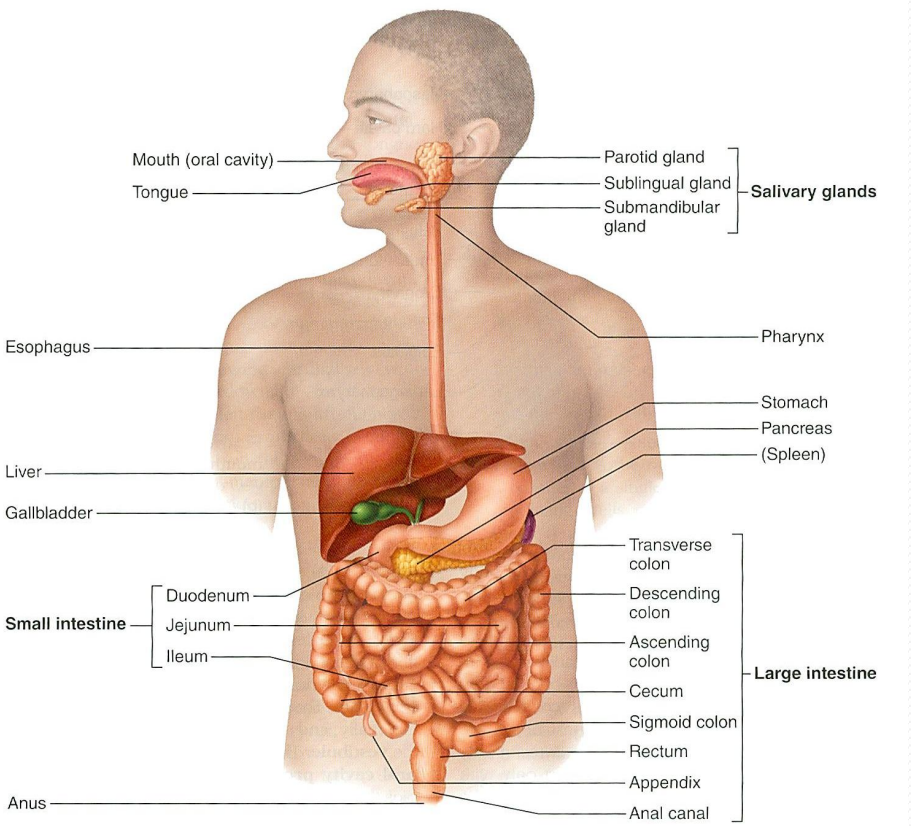
- Connective tissue and fascicles
  - Epimysium 肌外膜
    Surrounds the whole skeletal muscle
  - Perimysium 肌束膜
    Surrounds each fascicle
  - Endomysium 肌内膜
    Surrounds each muscle fiber
- Nerves and blood vessels



Connective tissue wrappings of skeletal muscle.

Digestive system

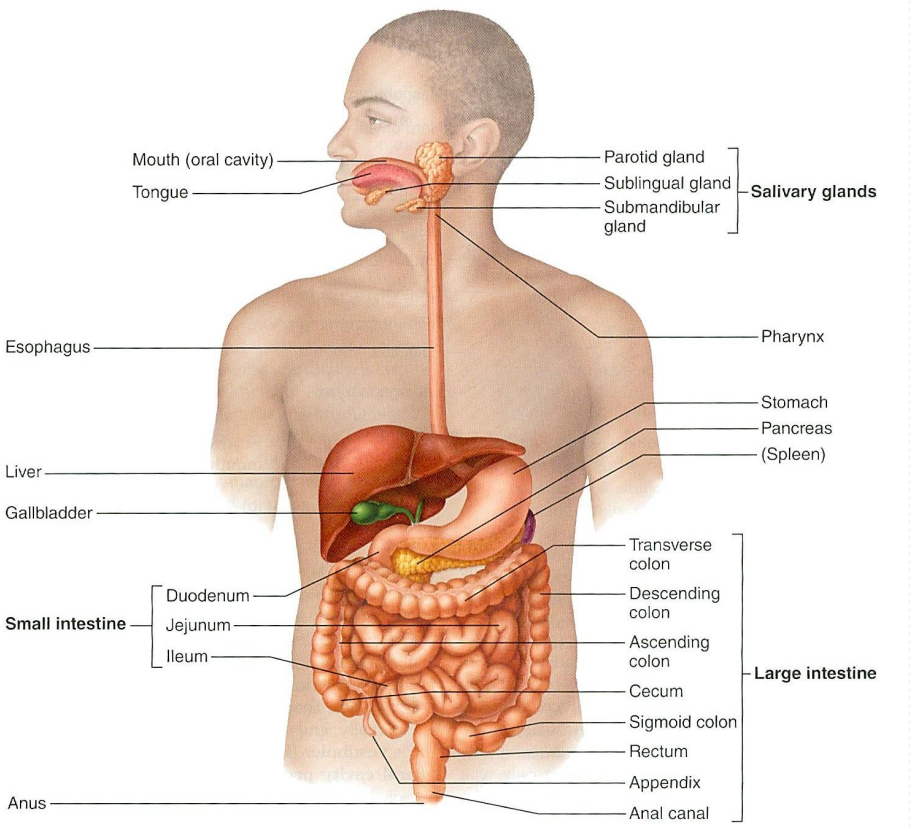
- **Alimentary canal 消化管**
  gastrointestinal (GI) tract
  - Mouth 口腔
  - Pharynx 咽
  - Esophagus 食管
  - Stomach 胃
  - Small intestine 小肠
  - Large intestine 大肠



The human digestive system: Alimentary canal and accessory organs.

Digestive system

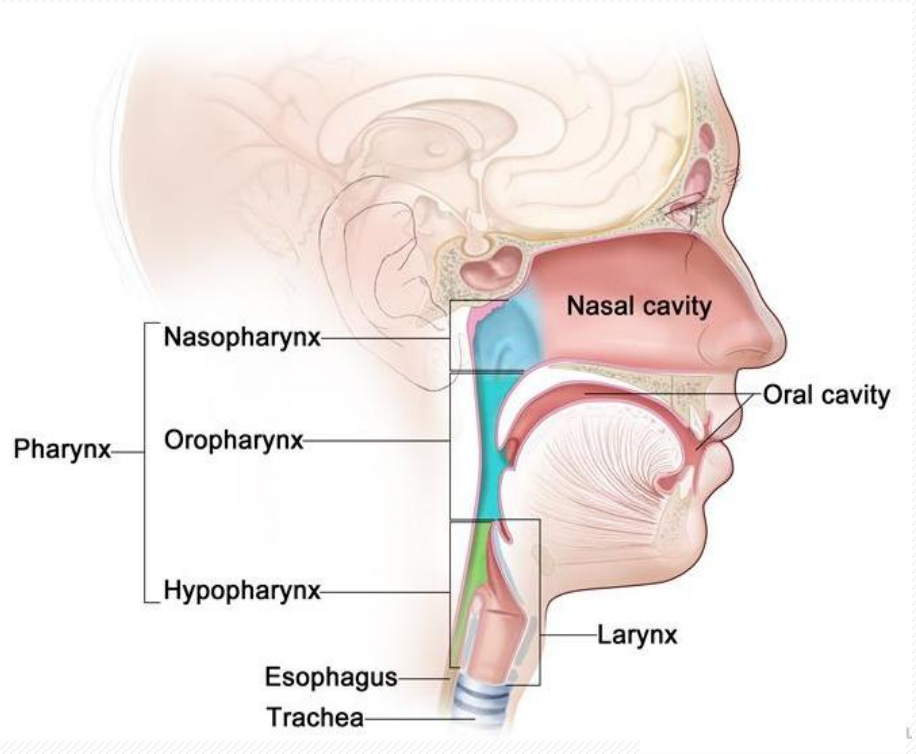
- **Accessory organs**
  Alimentary glands 消化腺
  - Major salivary glands
  - Liver
  - Pancreas



The human digestive system: Alimentary canal and accessory organs.

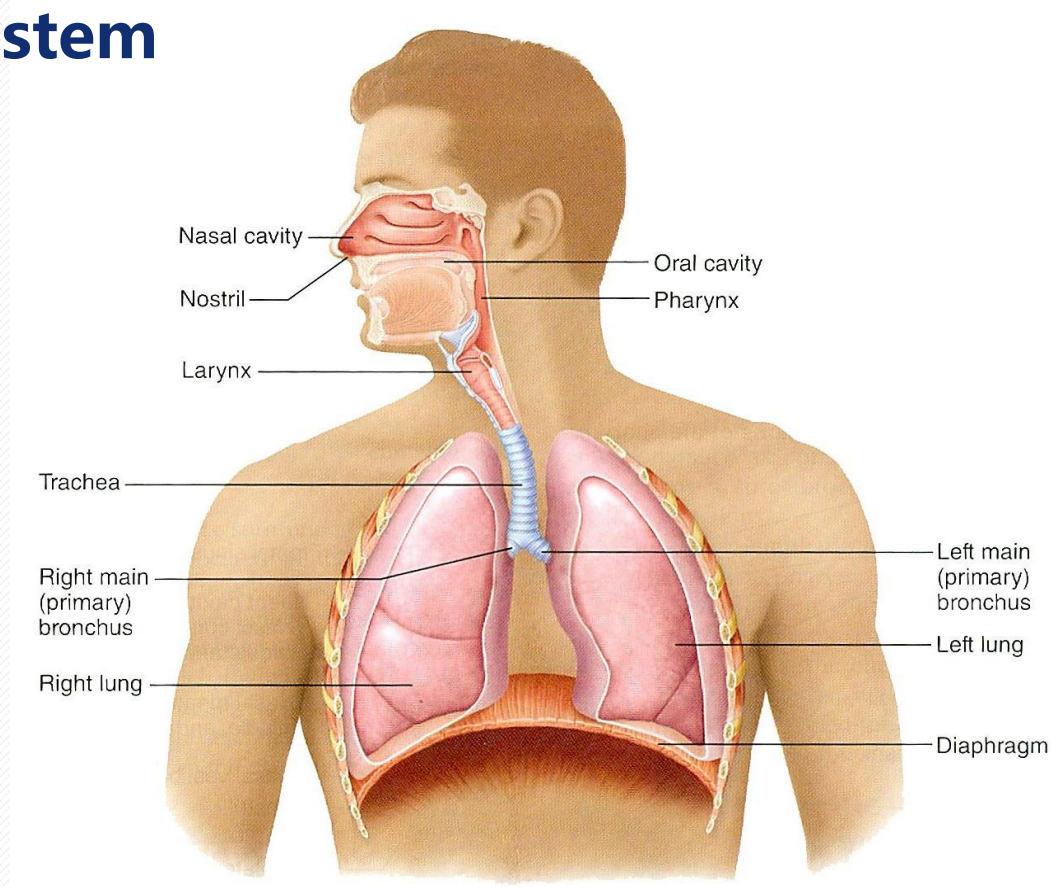
Pharynx

- Nasopharynx 鼻咽
- Oropharynx 口咽
- Laryngopharynx 喉咽



Respiratory system

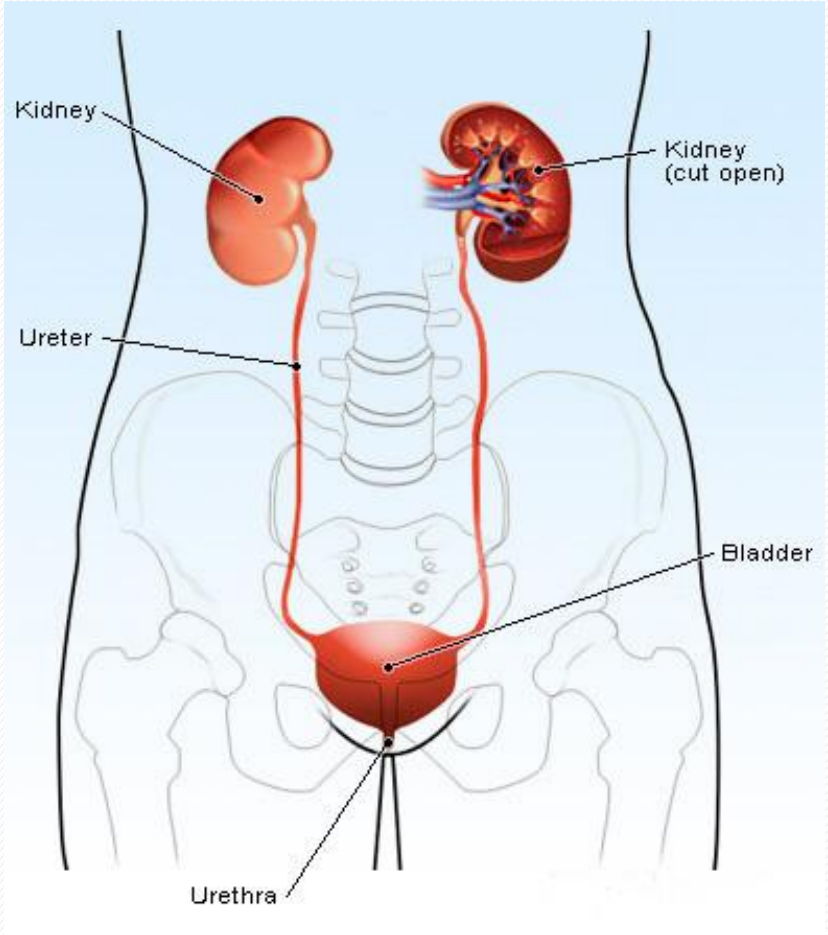
- Respiratory Tract
  - Nose 鼻
  - Pharynx 咽
  - Larynx 喉
  - Trachea 气管
  - Bronchi 支气管
- Lung



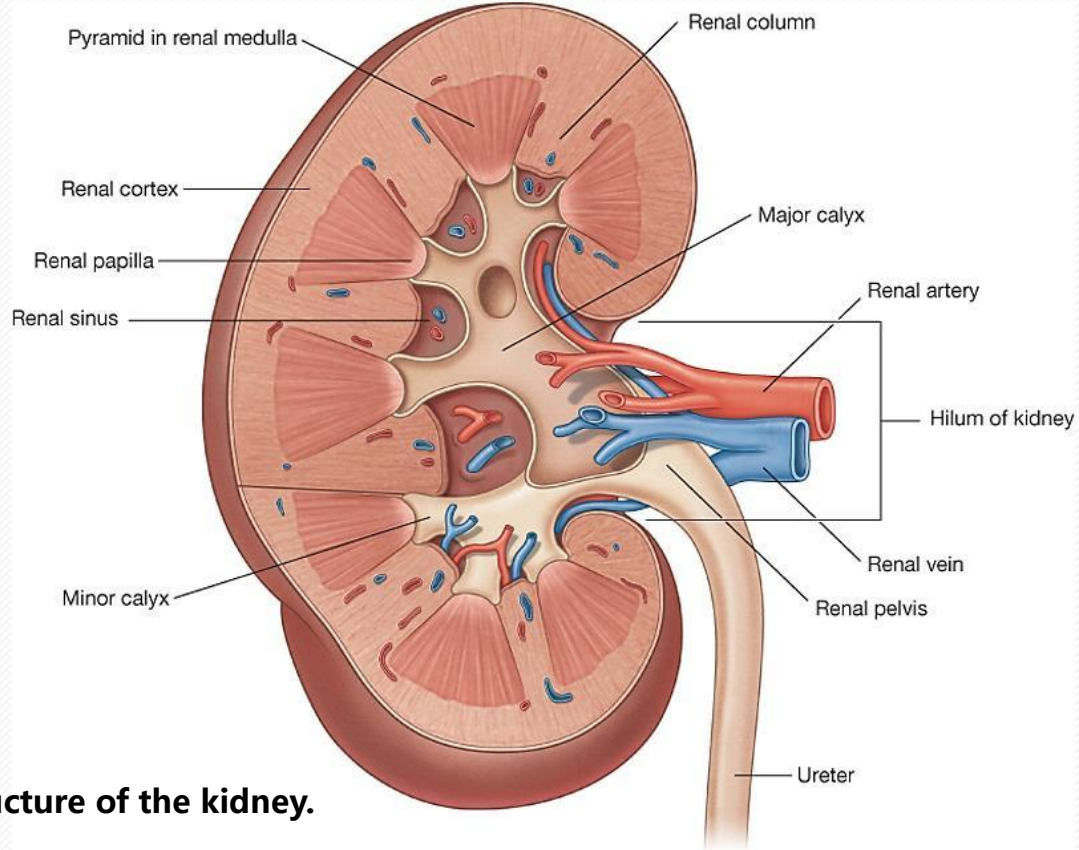
The major respiratory organs shown in relation to surrounding structures.

Urinary system

- Kidney 肾
- Ureter 输尿管
- Urinary bladder 膀胱
- Urethra 尿道



Kidney

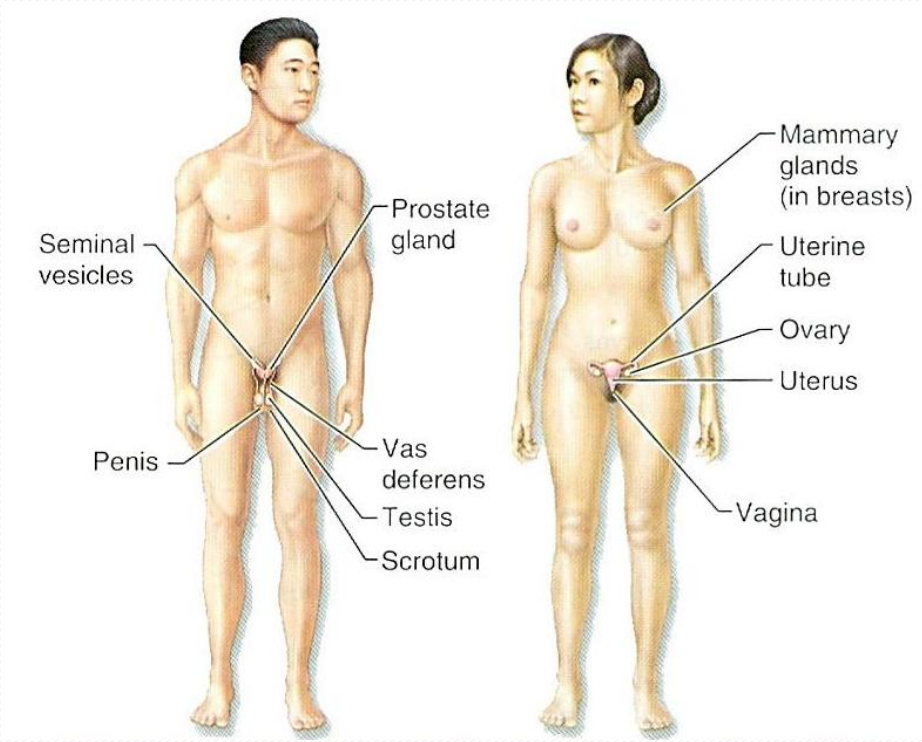


Internal structure of the kidney.

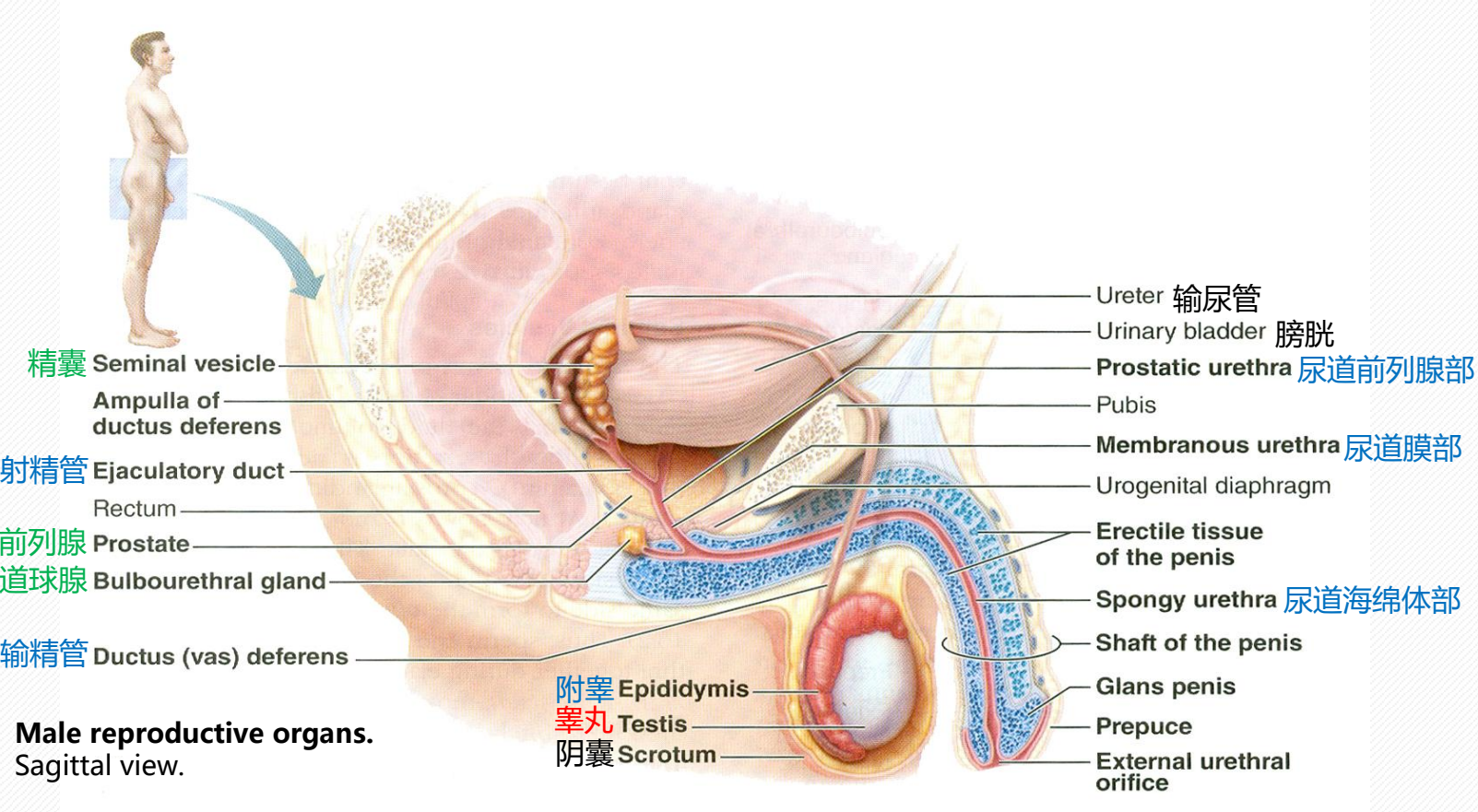
Reproductive system

Composition

- **Internal genital organ**
  - Gonads (sex glands)
    - manufacture the sex cells an secrete the sex hormones.
  - Genital ducts
    - transport the sex cells from the site of production into site of fertilization
  - Accessory glands
    - secrete the fluid
- **External genital organ**

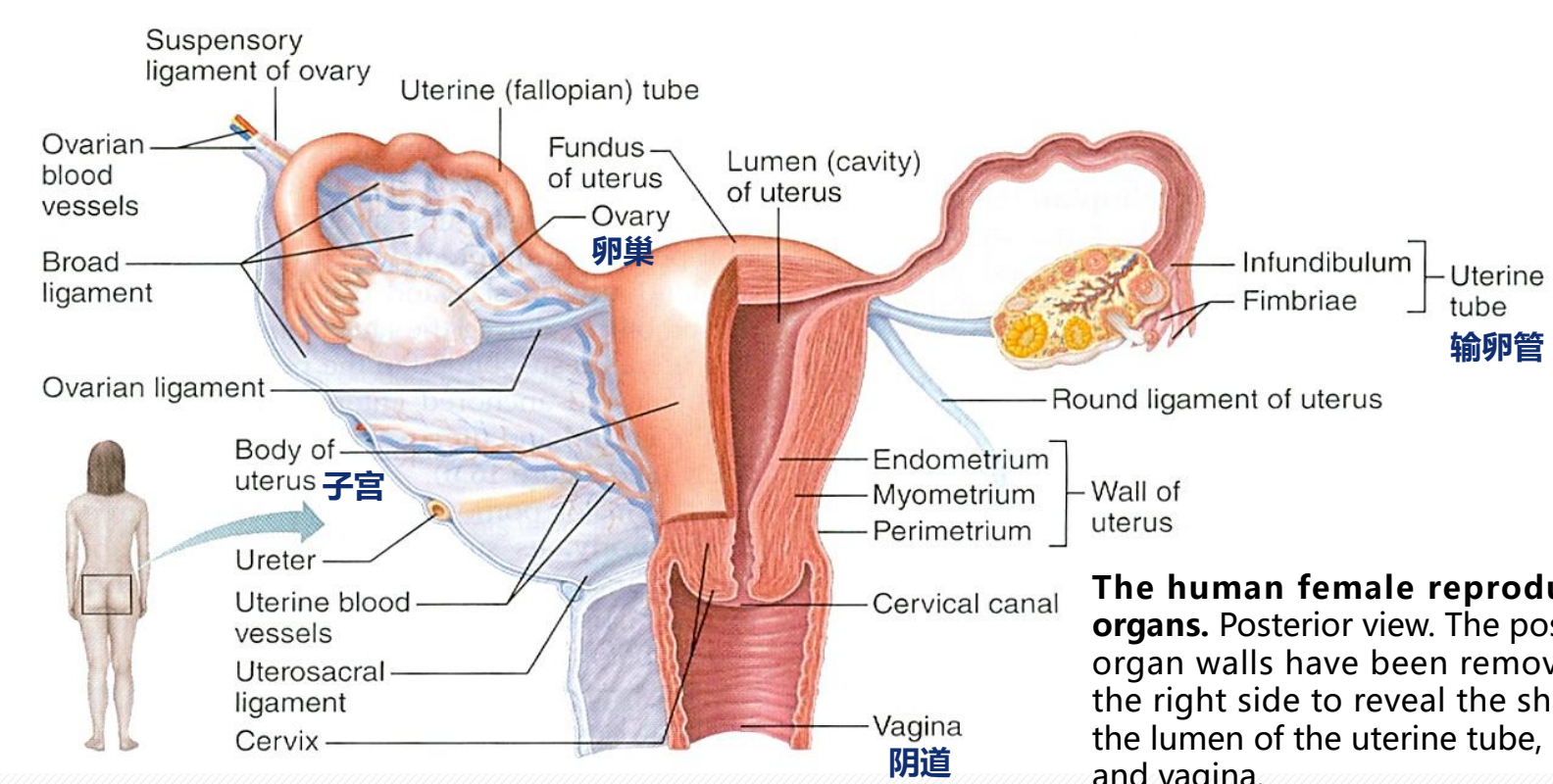


Male reproductive system



Male reproductive organs.
Sagittal view.

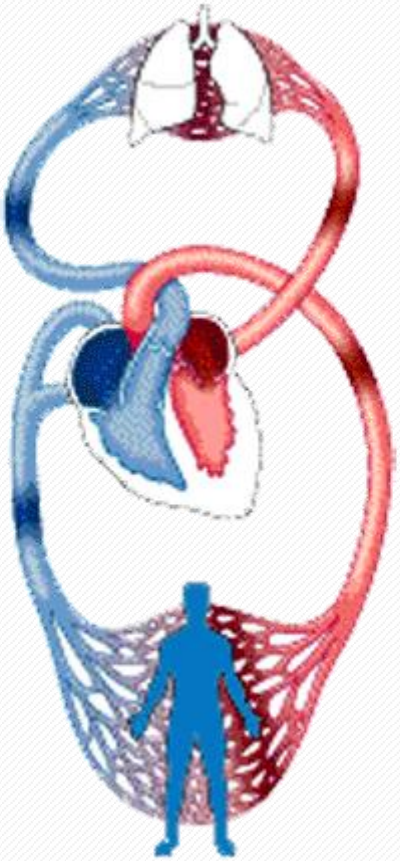
Female reproductive system



The human female reproductive organs. Posterior view. The posterior organ walls have been removed on the right side to reveal the shape of the lumen of the uterine tube, uterus, and vagina.

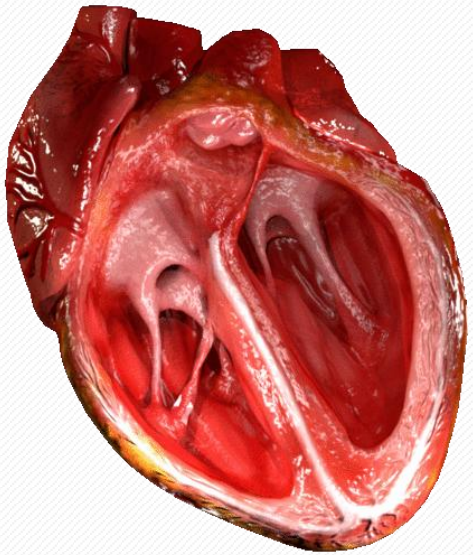
Cardiovascular system

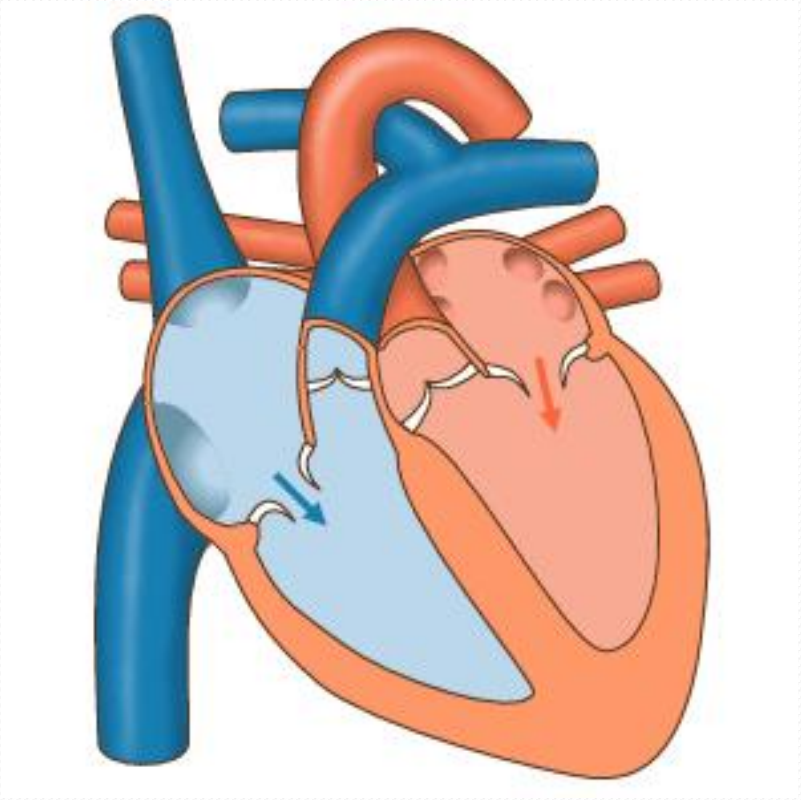
The systemic and pulmonary circulations



The lower side of the heart is the systemic pump; the upper side is the pulmonary circuit pump.

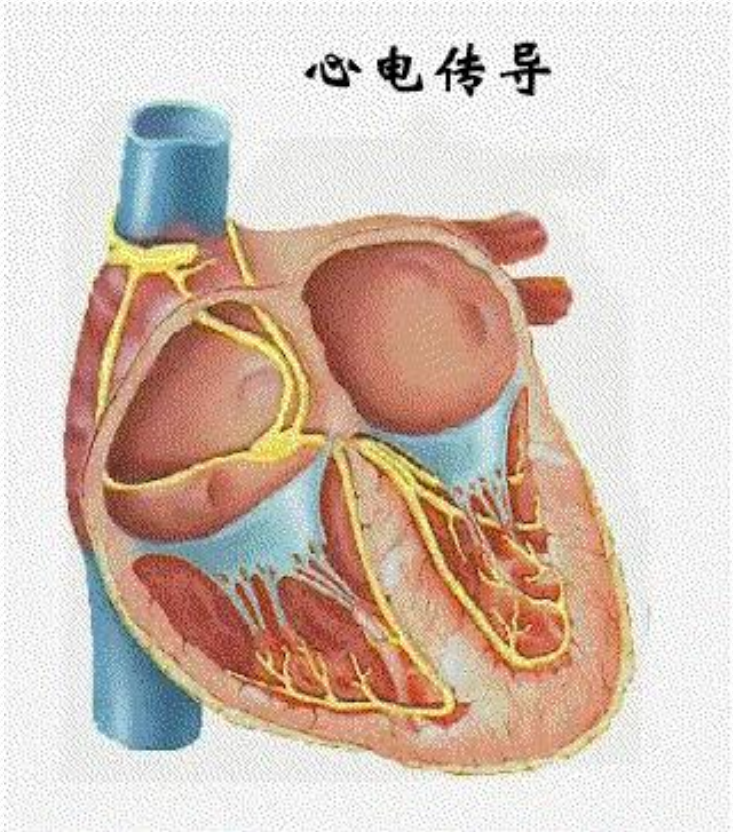
Heart

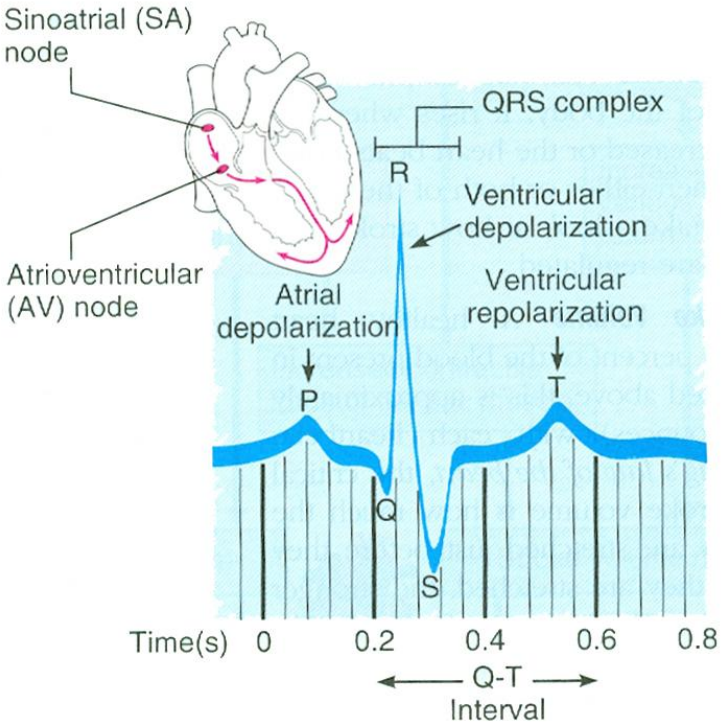




Blood flow through the valves

Heart





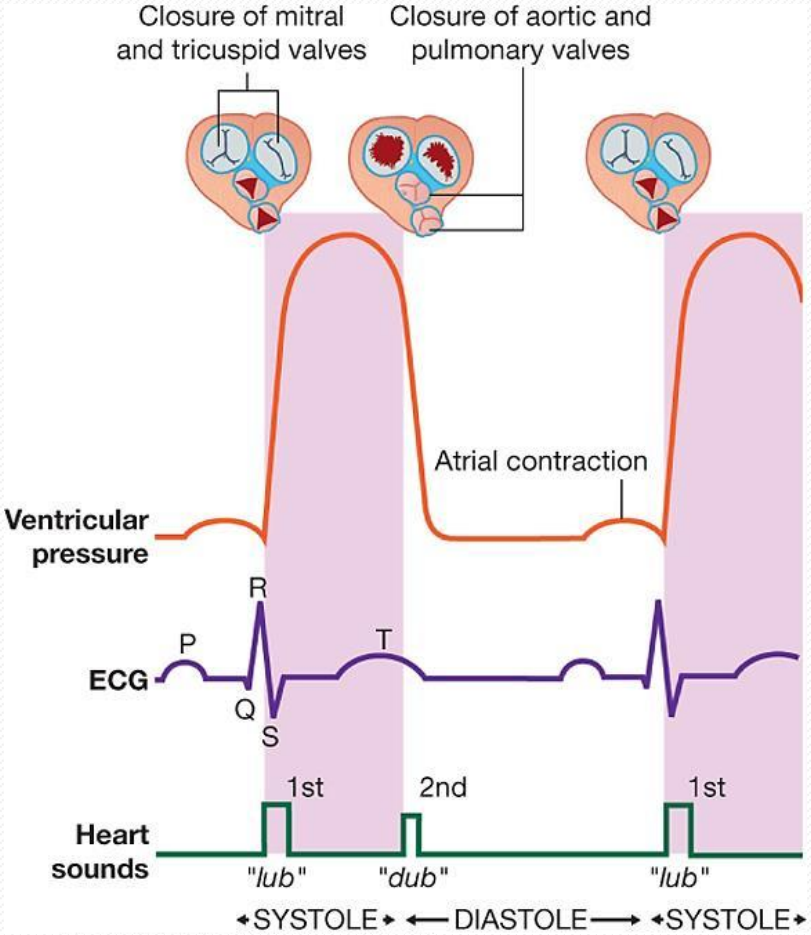
An electrocardiogram tracing showing the three normally recognizable deflection waves—P, QRS, and T.

Heart

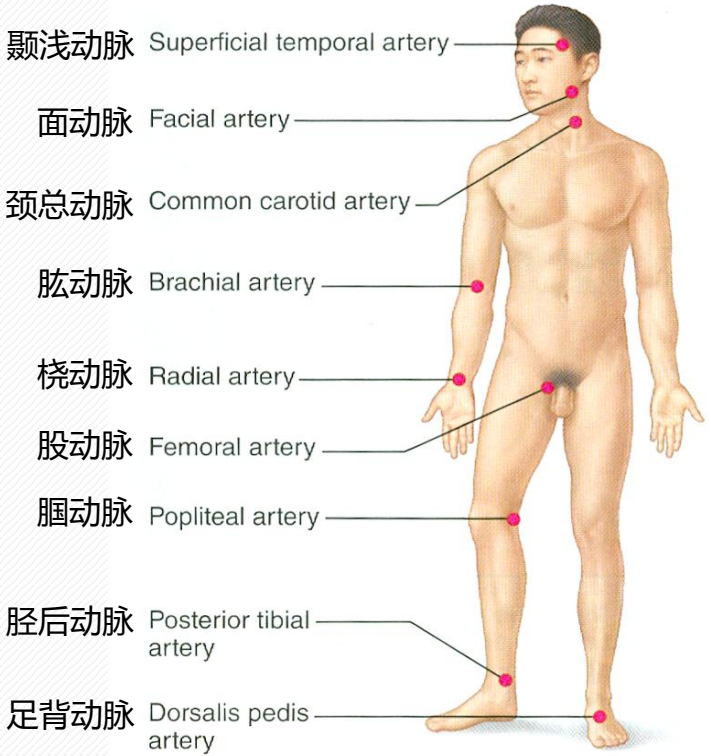
Heart sounds

- The first heart sound is caused by the closing of the AV valves.
- The second heart sound occurs when the semilunar valves close at the end of systole.

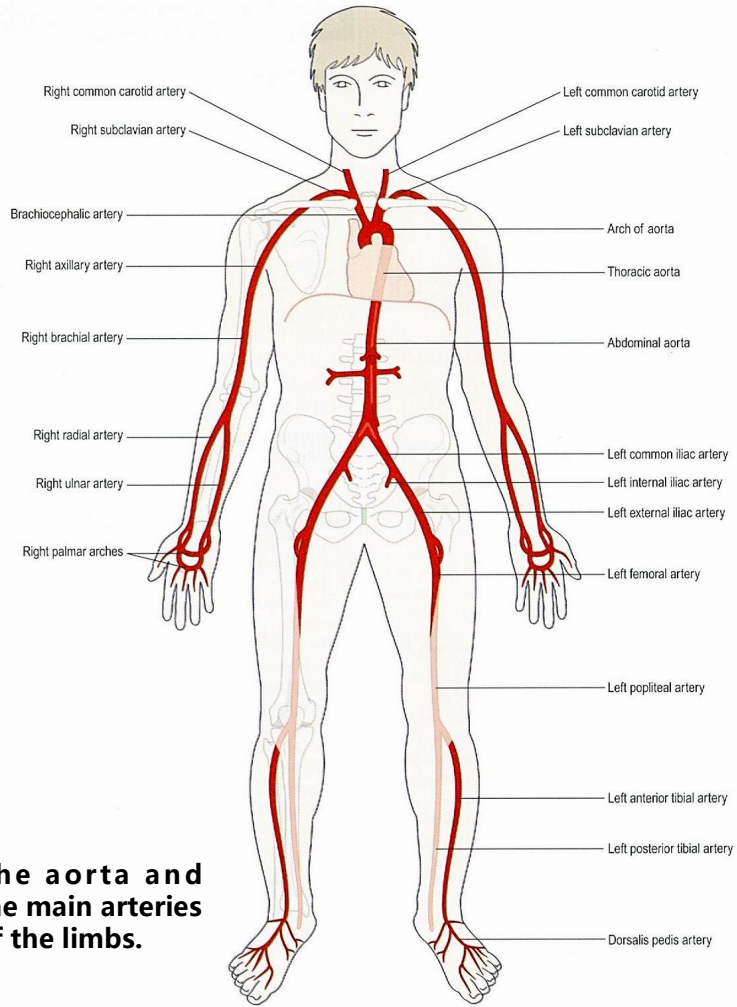
Heart sounds and how they relate to valve closure, the electrocardiogram (ECG), and ventricular pressure.



Arteries



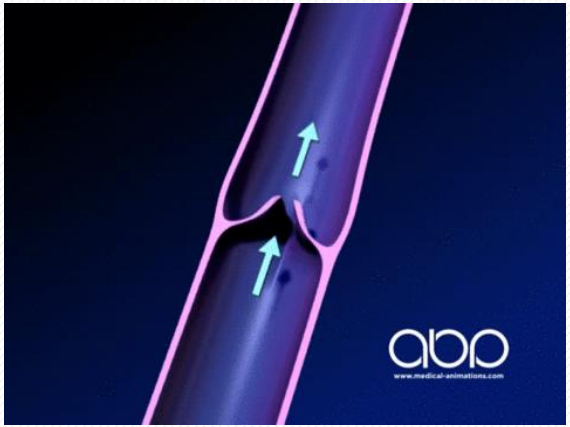
Body sites where the pulse is most easily palpated.

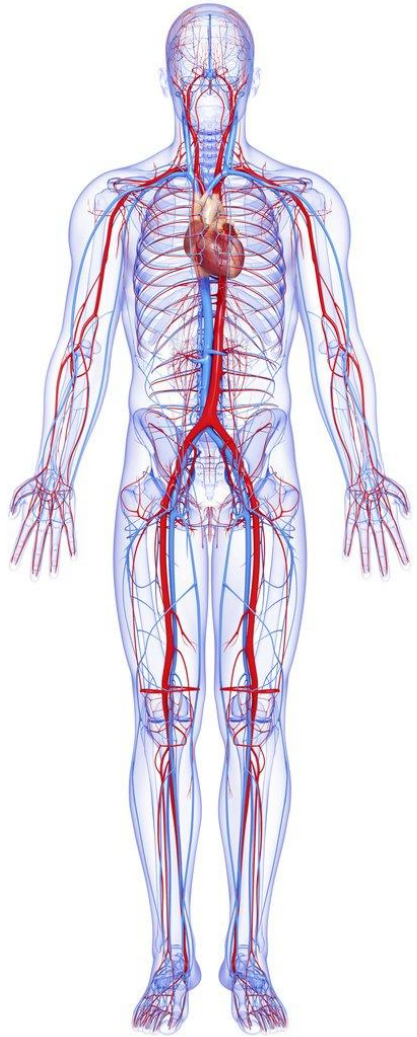


The aorta and the main arteries of the limbs.

Veins

- Superficial veins
- Deep veins
  --Venous valve

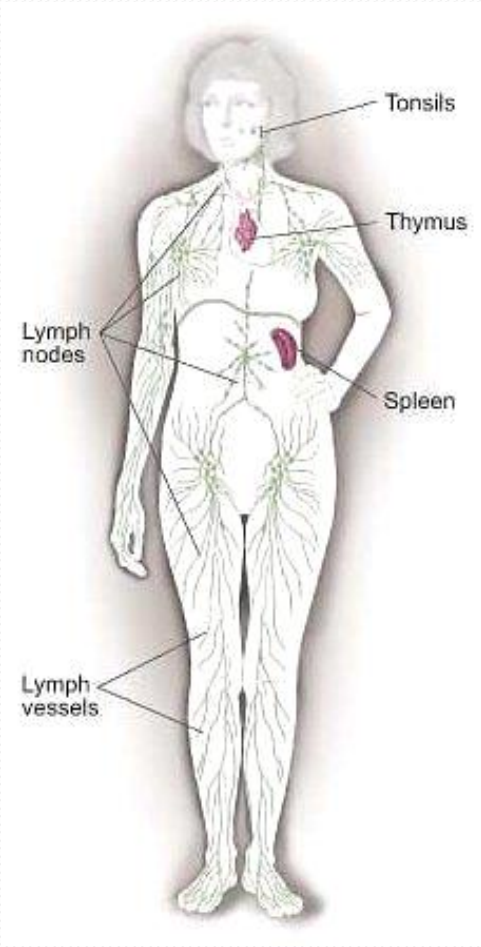




Lymphatic system

- It returns leaked plasma to the blood vessels after cleansing it of bacteria and other foreign matter.
- It also provides sites for surveillance by immune system cells.

- **Lymphatic vessels**
- **Lymphoid tissues and organs**

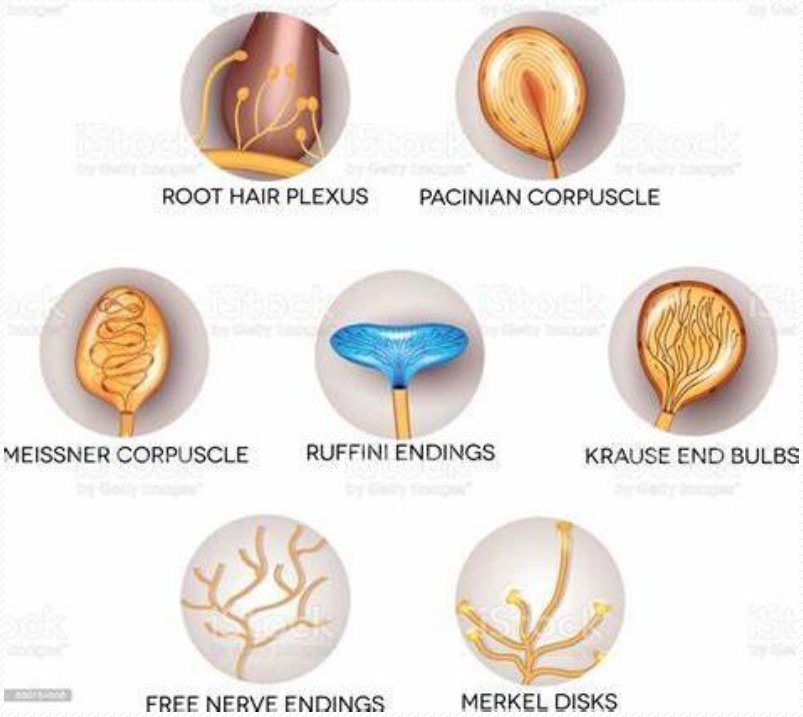


Sensory receptors

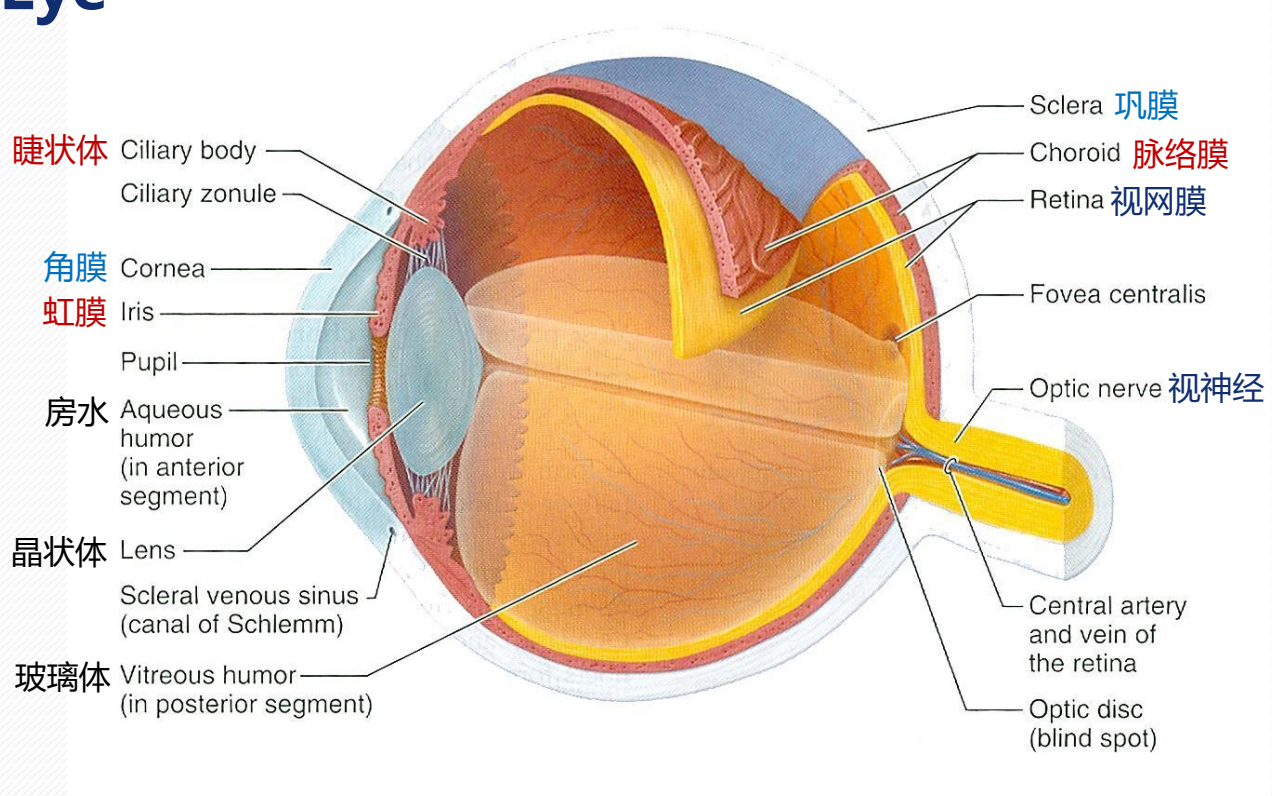
Types of Sensory Receptors based on Structure

- **Free nerve endings or dendrites**
  – E.g. thermoreceptors, nociceptors
- **Encapsulated nerve endings**
  - E.g. mechanoreceptors
- **Specialized receptor cells**
  - E.g. photoreceptors.

Sensory receptors in the skin

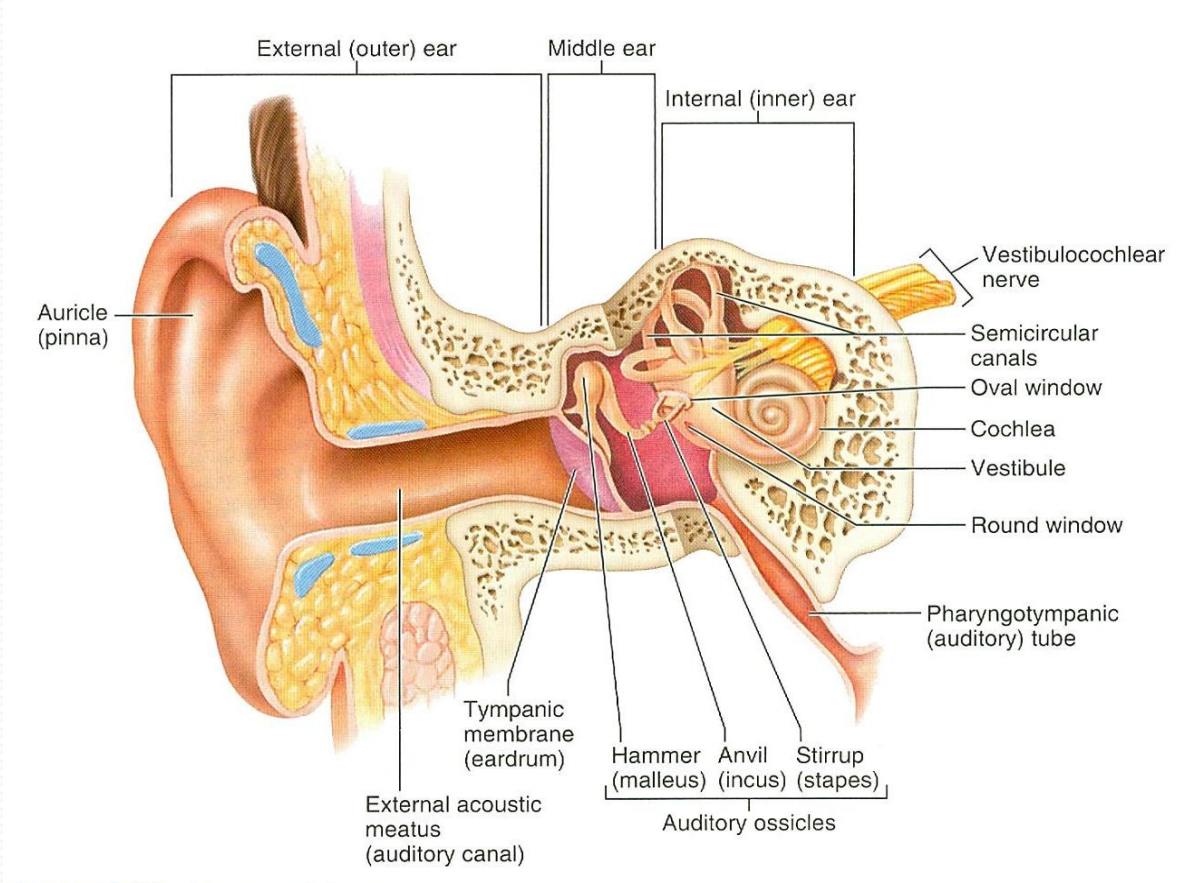


Eye

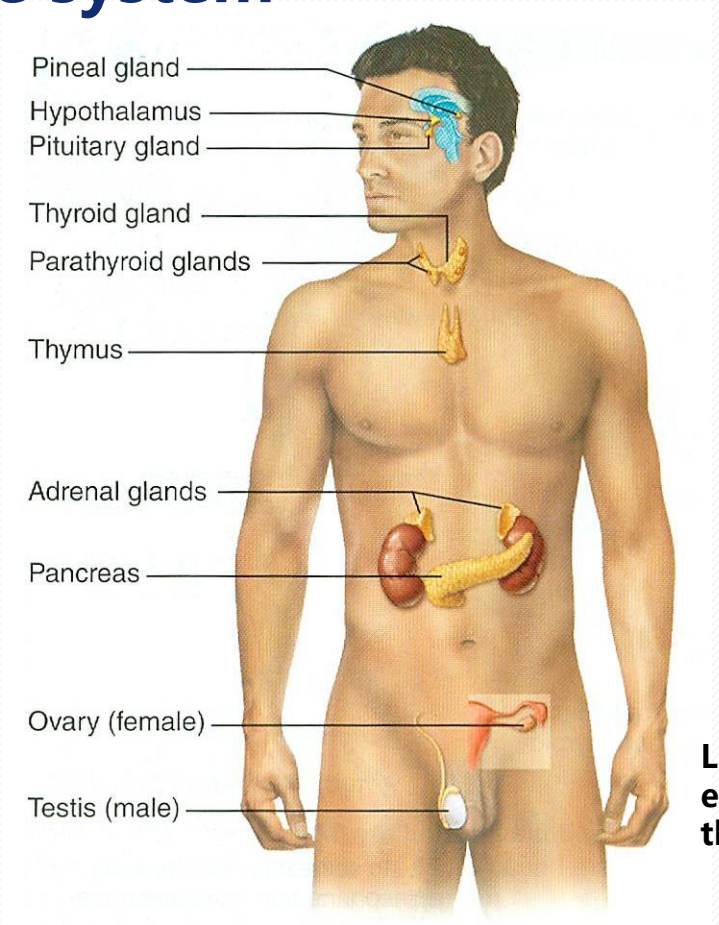


Internal anatomy of the eye (sagittal section).

Ear


External (outer) ear
Middle ear
Internal (inner) ear
Auricle (pinna)
Vestibulocochlear nerve
Semicircular canals
Oval window
Cochlea
Vestibule
Round window
Pharyngotympanic (auditory) tube
Tympanic membrane (eardrum)
Hammer (malleus)
Anvil (incus)
Stirrup (stapes)
Auditory ossicles
External acoustic meatus (auditory canal)

Endocrine system



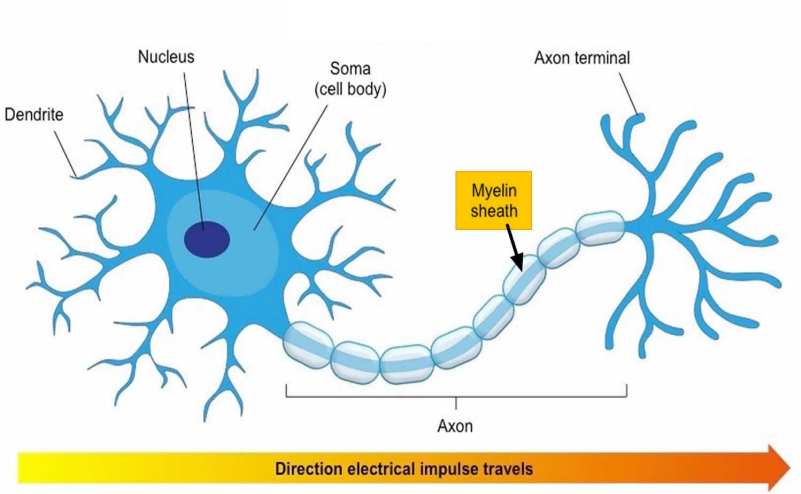
Location of the major endocrine organs of the body.

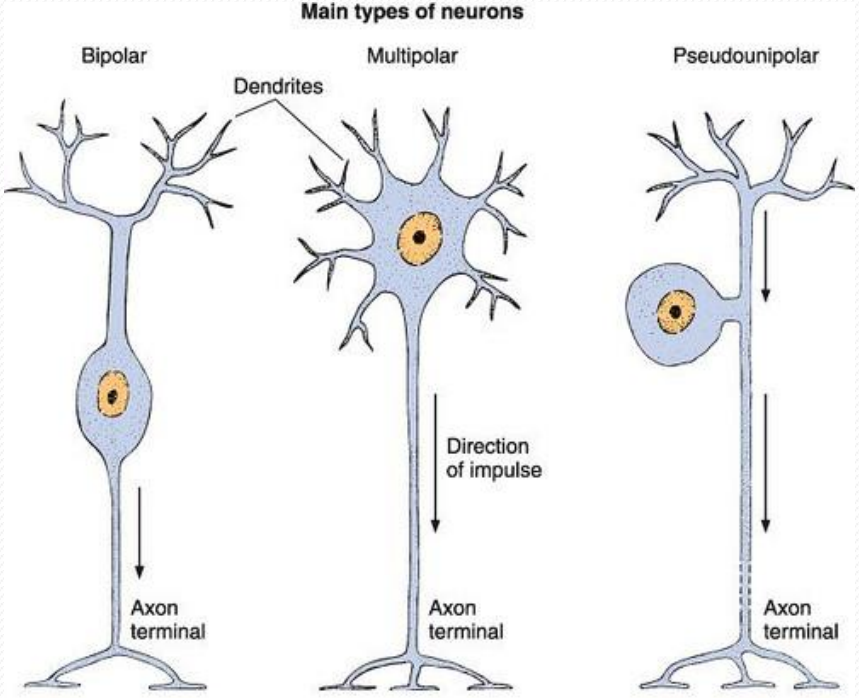
Part II

Neuroanatomy

Neural tissue

Neuron



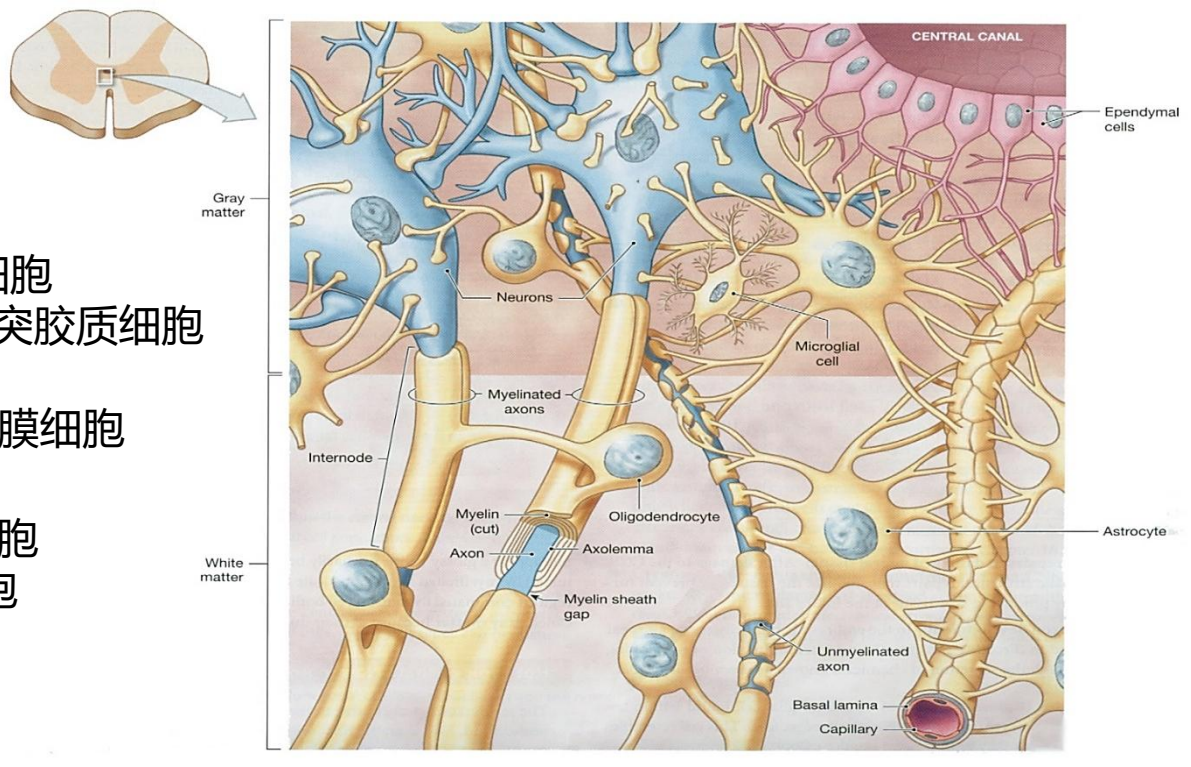


Classification of neurons on the basis of structure.

Neural tissue

Neuroglia

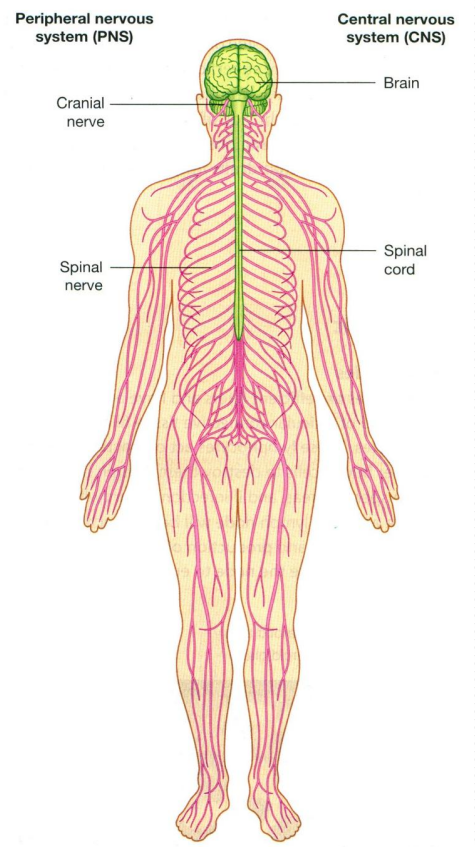
- CNS
  - Astrocytes 星形胶质细胞
  - Oligodendrocytes 少突胶质细胞
  - Microglia 小胶质细胞
  - Ependymal cells 室管膜细胞
- PNS
  - Schwann cells 施万细胞
  - Satellite cells 卫星细胞



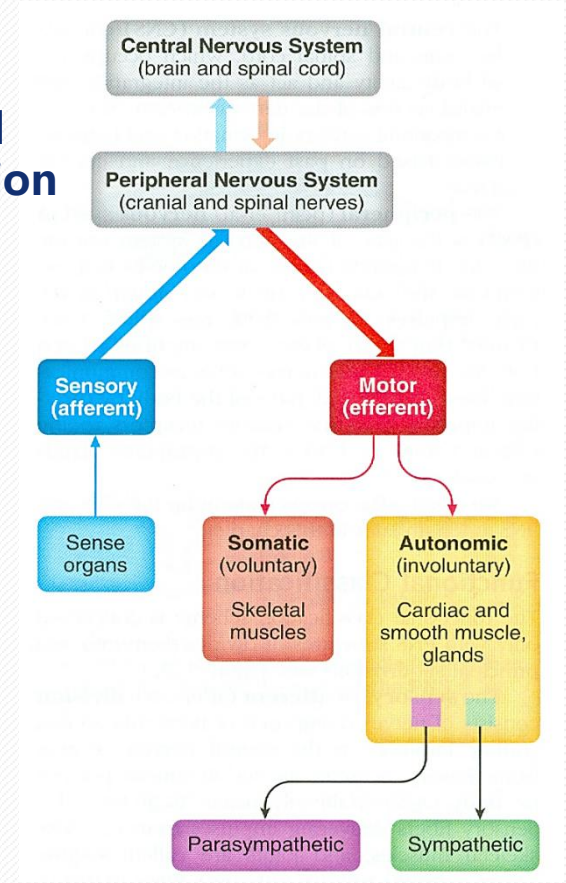
Histology of Neural Tissue in the CNS.
A diagrammatic view of neural tissue in the spinal cord, showing relationships between neurons and glial cells.

Nervous system

Structural classification



Functional classification



Nervous system

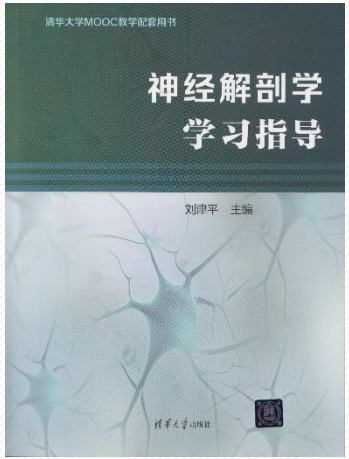
- **Peripheral nervous system (PNS)**
  - Spinal nerves 脊神经
  - Cranial nerves 脑神经
  - Visceral nervous system 内脏神经
- **Central nervous system (CNS)**
  - Spinal cord 脊髓
  - Brain
    - Brainstem 脑干
    - Cerebellum 小脑
    - Diencephalon 间脑
    - Telencephalon 端脑

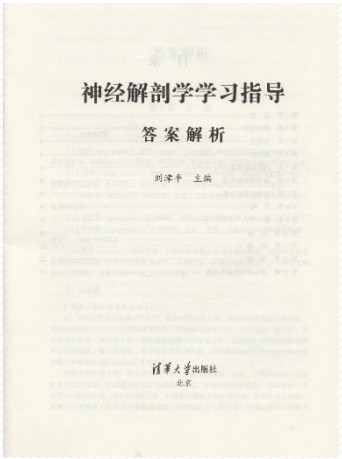


神经解剖学--周围神经系统
清华大学
刘津平
长按识别二维码

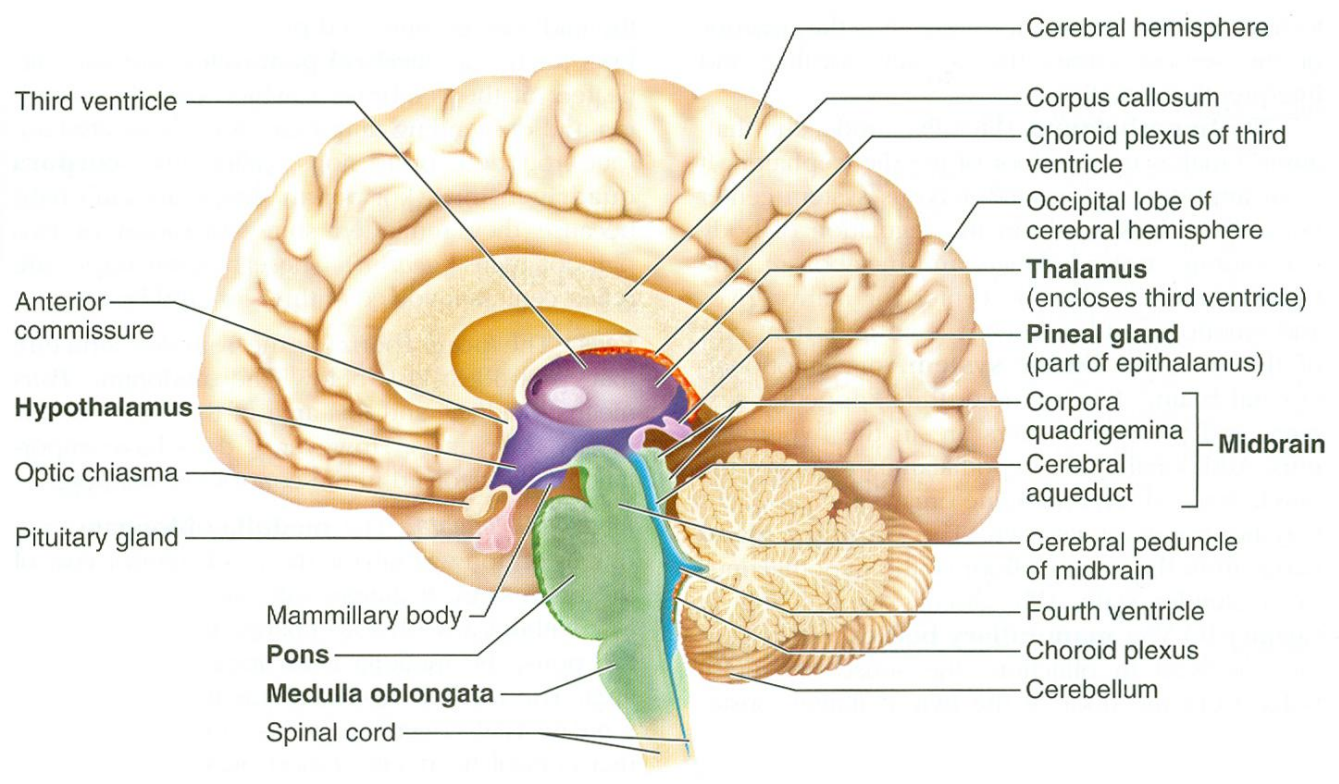


神经解剖学--中枢神经系统
清华大学
刘津平
长按识别二维码



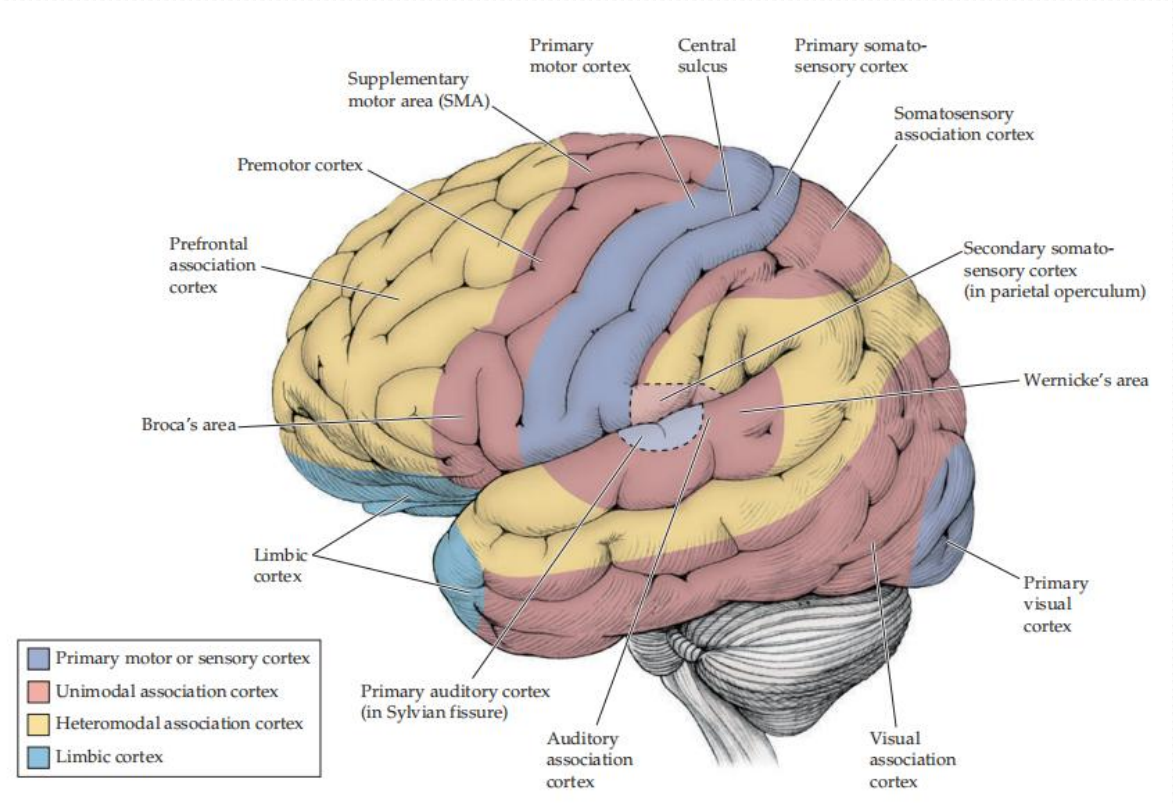


Nervous system



Midsagittal section of the brain.

Telencephalon



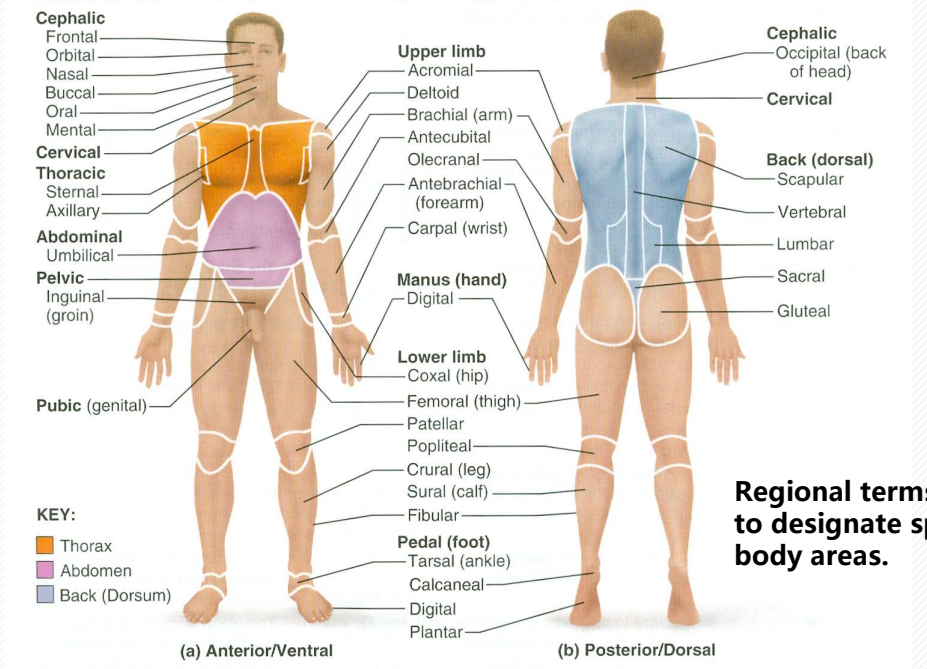
Association cortex. Lateral view of left hemisphere showing main areas of primary sensory and motor cortex, unimodal association cortex, and heteromodal association cortex.

Part III Topographic anatomy

Topographic anatomy

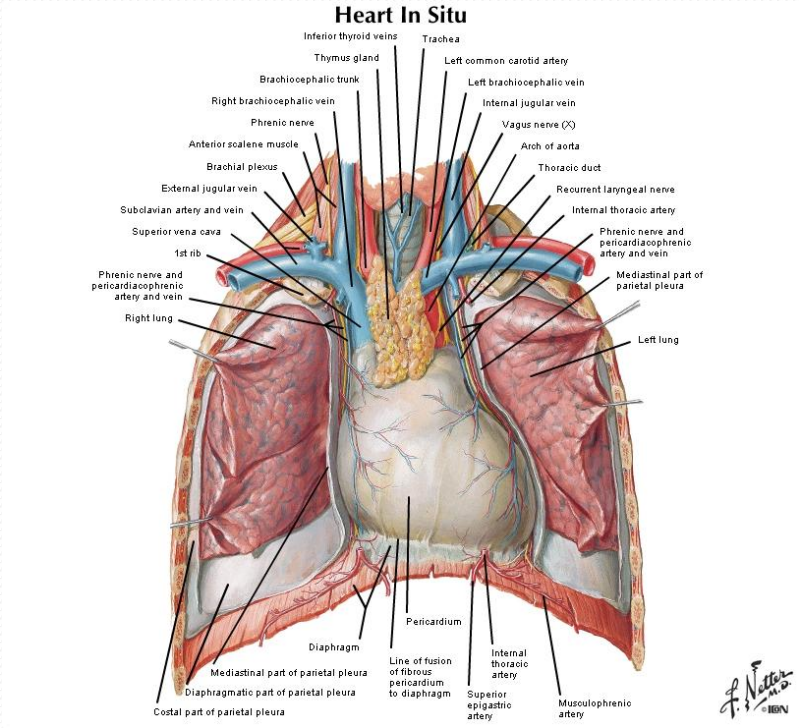
The study of anatomy by regional parts of the body, e.g., thorax, lower limb, etc. Under this mode of study all of the biological systems, e.g., skeletal, circulatory, etc. located within a particular body region are simultaneously studied with an emphasis on the interrelation of the systems and their regional function.

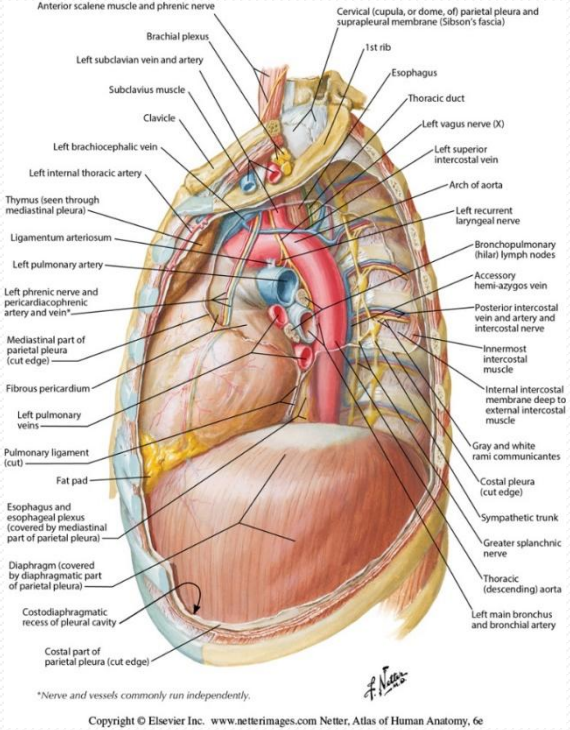
- Head
- Neck
- Back
- Thorax 胸部
- Abdomen 腹部
- Pelvis 盆部
- Perineum 会阴部
- Upper limbs
- Lower limbs



Regional terms used to designate specific body areas.

Thorax


Heart In Situ
Inferior thyroid veins
Trachea
Thymus gland
Left common carotid artery
Brachiocephalic trunk
Left brachiocephalic vein
Right brachiocephalic vein
Internal jugular vein
Phrenic nerve
Vagus nerve (X)
Anterior scalene muscle
Arch of aorta
Brachial plexus
Thoracic duct
External jugular vein
Recurrent laryngeal nerve
Subclavian artery and vein
Internal thoracic artery
Superior vena cava
Phrenic nerve and pericardiacophrenic artery and vein
1st rib
Phrenic nerve and pericardiacophrenic artery and vein
Mediastinal part of parietal pleura
Right lung
Left lung
Pericardium
Diaphragm
Mediastinal part of parietal pleura
Line of fusion of fibrous pericardium to diaphragm
Internal thoracic artery
Diaphragmatic part of parietal pleura
Superior epigastric artery
Musculophrenic artery
Costal part of parietal pleura


Anterior scalene muscle and phrenic nerve
Cervical (cupula, or dome, of) parietal pleura and suprapleural membrane (Sibson's fascia)
Brachial plexus
1st rib
Left subclavian vein and artery
Esophagus
Subclavius muscle
Thoracic duct
Clavicle
Left vagus nerve (X)
Left brachiocephalic vein
Left superior intercostal vein
Left internal thoracic artery
Arch of aorta
Thymus (seen through mediastinal pleura)
Left recurrent laryngeal nerve
Ligamentum arteriosum
Bronchopulmonary (hilar) lymph nodes
Left pulmonary artery
Accessory hemi-azygos vein
Left phrenic nerve and pericardiacophrenic artery and vein*
Posterior intercostal vein and artery and intercostal nerve
Mediastinal part of parietal pleura (cut edge)
Innermost intercostal muscle
Fibrous pericardium
Internal intercostal membrane deep to external intercostal muscle
Left pulmonary veins
Gray and white rami communicantes
Pulmonary ligament (cut)
Costal pleura (cut edge)
Fat pad
Sympathetic trunk
Esophagus and esophageal plexus (covered by mediastinal part of parietal pleura)
Greater splanchnic nerve
Diaphragm (covered by diaphragmatic part of parietal pleura)
Thoracic (descending) aorta
Costodiaphragmatic recess of pleural cavity
Left main bronchus and bronchial artery
Costal part of parietal pleura (cut edge)
*Nerve and vessels commonly run independently.
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Thanks for your attention

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